
Long-Range, Off-Grid Emergency Alert Messaging System Using LoRa Technology

Abhijeet Arun Dalvi, Atharv Ranveer Pawar, Bhushan Sandeep Todkar

Under the Guidance of Prof. S. L. Tatuskar

Department of Electronics and Telecommunication Engineering, Modern College of Engineering, Pune, India
Savitribai Phule Pune University | Academic Year: 2025-26

ABSTRACT- Disaster Management leads to address various issues like floods, cyclones, fire, etc. Different regions are vulnerable to different disasters due to geo-climatic conditions and varied landscapes. This project addresses this critical need through a LoRa-based long-range, off-grid emergency communication system using LoRa SX1278 modules (433 MHz) and STM32 Blue Pill (STM32F103C8T6) microcontrollers. It is designed for extremely affordable and minimal power needs. It allows victims to broadcast 'HELP' or 'SAFE' messages with GPS coordinates to stationary receiver nodes up to 10 km away. By restoring basic communication, the system bridges the deadly silence between victims and rescuers when every minute counts. The receiver displays decoded messages on an SSD1306 OLED screen. Since the system operates without SIM cards or internet infrastructure, it is highly suitable for disaster management, trekking, and rescue operations.

Keywords- LoRa, SX1278, STM32 Blue Pill, Emergency Communication, Off-Grid, GPS, OLED, Disaster Management, Chirp Spread Spectrum.

I. INTRODUCTION

In recent years, reliable communication has emerged as a critical requirement — especially in emergency situations such as natural disasters, remote area operations, and search-and-rescue missions. Conventional communication systems such as cellular networks and internet-based services frequently fail during such scenarios due to infrastructure damage or the absence of coverage in remote terrain. This breakdown creates a significant information gap, delaying rescue operations and ultimately costing lives.

Natural and man-made disasters frequently cripple conventional communication networks, leaving affected populations isolated when they need help most. Earthquakes, tsunamis, floods, and cyclones regularly devastate regions across the globe, while even developed nations face vulnerabilities — Hurricane Katrina (2005) disabled 70% of cell towers in New Orleans, and the 2011 Tōhoku earthquake disrupted Japan's communication infrastructure for weeks [1]. These breakdowns create deadly information gaps where victims cannot call for help and responders lack situational awareness. Traditional radios like walkie-talkies have limited range, while satellite phones remain prohibitively expensive for widespread deployment.

Unlike cellular networks, LoRa's long-range, low-bandwidth protocol penetrates dense urban rubble and remote terrain. Field tests in simulated disaster zones have demonstrated up to 92% message reliability despite obstructions [1]. The motivations for adopting LoRa-based communication include: 1) low cost, 2) acceptable communication range, 3) low energy consumption, and 4) high possibility of deployment in remote areas where the internet and cellular networks do not exist.

The primary objective of this project is to develop a long-range, off-grid communication system using LoRa technology that functions entirely independently of existing communication infrastructure. The system transmits predefined emergency messages over long distances with minimal power consumption, ensuring reliable connectivity even in the most challenging field environments.

1.1 LoRaWAN and LoRa Technology

The LoRaWAN protocol is a Low Power Wide Area Network (LPWAN) designed for wireless connection of equipment to the

internet [5], achieving bi-directional communication between nodes and servers. The gateway functions like a router that relays data from various nodes to the network server.

LoRa is the physical layer wireless modulation used to create long-range communication links [5]. Most known wireless technologies use Frequency Shift Keying (FSK) modulation for low-power communication. LoRa, on the other hand, uses Chirp Spread Spectrum (CSS) modulation, which achieves both low power and long-range communication. CSS works with chirps whose frequencies increase or decrease linearly over time [5]. LoRa operates in the free ISM spectrum band — 433 MHz (Asia), 868 MHz (Europe), and 915 MHz (USA) [5].

Four important parameters control LoRa communication: 1) Transmission Power (typically 2–20 dBm), 2) Spreading Factor (SF6–SF12), 3) Bandwidth (125/250/500 kHz), and 4) Coding Rate — the amount of forward error correction (FEC) applied [10]. The transmission rate is given by:

$$TR = SF \times BW / 2^{SF} \times CR \dots (1)$$

A LoRa transceiver can decode transmissions up to 20 dB below the noise floor, making very long-distance low-power communication practical.

1.2 LoRa Software Implementation

LoRa communication is implemented using open-source code libraries available on GitHub, configured for microcontrollers via C/C++. The nodes send processed data to the gateway, through which data is forwarded to a network server. This open-source ecosystem significantly contributes to the low cost of LoRa deployments.

II. LITERATURE SURVEY

A growing body of research has explored LoRa and LoRaWAN technology in emergency and IoT communication contexts. Zhou et al. [2] examined the design and implementation of Open LoRa for IoT, demonstrating the protocol's versatility across diverse deployment scenarios. Blesson [3] developed a LoRa-based emergency communication system, highlighting its suitability for low-power alert messaging over extended ranges.

Mahakal et al. [4] proposed a cordless communication system using LoRa technology that demonstrated the viability of infrastructure-independent communication for civilian use. Murugan et al. [1] specifically addressed disaster management with a simplex LoRa communication system using Arduino and LoRa RA-02 modules, allowing victims to broadcast SOS messages with GPS coordinates to stationary receiver nodes up to 10 km away.

Pasolini et al. [6] investigated Smart City applications of LoRa and IEEE 802.15.4 technologies, while Aoudia et al. [7] explored long-short range networks leveraging LoRa with wake-up receivers. Abhishek IJ [8] proposed a LoRa-based emergency device capable of operating without any external power or network infrastructure.

From the literature survey, a key observation emerges: most existing implementations focus on IoT data logging or large-scale networks, leaving a gap in simple, portable, button-based emergency alert systems targeting non-technical end users. The present work addresses this gap with a point-to-point STM32-based design featuring GPS integration and OLED output.

III. PROBLEM STATEMENT AND OBJECTIVES

A. Problem Statement

In remote and disaster-affected areas, reliable communication becomes a major challenge due to the failure or total absence of conventional communication infrastructure. This creates critical difficulty in sending emergency information, leading to delays in rescue operations.

B. Objectives

The main objectives of this project are:

- To design and implement a wireless communication system using LoRa SX1278 and STM32 microcontrollers.
- To enable long-range transmission of emergency messages without cellular networks or internet infrastructure.
- To develop a simple, button-based user interface for sending predefined alert messages.
- To minimize power consumption for extended battery-operated field deployment.
- To integrate GPS location data into transmitted emergency messages.
- To demonstrate reliable communication under non-line-of-sight (NLOS) conditions.

IV. SYSTEM ARCHITECTURE AND DESIGN

A. System Overview

The proposed system consists of two nodes — a transmitter (Node 1) and a receiver (Node 2) — both built around the STM32 Blue Pill (STM32F103C8T6) microcontroller, communicating via LoRa SX1278 modules at 433 MHz. The system operates on a point-to-point communication model, entirely independent of cellular networks or internet connectivity.

Signal transmission follows a simple path: a button press on the transmitter encodes the selected emergency message, which is broadcast via the LoRa module. The receiver captures the packet, decodes it, and displays the message on an OLED screen in real time.

B. Node 1 – Transmitter

Node 1 integrates the following components and pin assignments:

- STM32F103C8T6 (Blue Pill): Central controller at 72 MHz, 3.3 V.

- LoRa SX1278: SPI interface — PA4 (NSS), PA5 (SCK), PA6 (MISO), PA7 (MOSI), PB0 (DIO0).
- HELP Button: PA0 + GND; SAFE Button: PB1 + GND.
- LED Indicators: PA8 (HELP LED), PA9 (SAFE LED) — 220 Ω current-limiting resistors.
- GPS NEO-6M: UART2 — PA3 (GPS RX), one-way NMEA data read.

C. Node 2 – Receiver

- STM32F103C8T6 (Blue Pill): Processes received LoRa packets and drives the OLED.
- LoRa SX1278: Identical SPI configuration to Node 1; operates in continuous receive mode.
- OLED SSD1306 (128×64): I2C interface — PB6 (SCL), PB7 (SDA); displays decoded messages.
- Buttons: PA0 (HELP), PA2 (SAFE) for local input / testing.

D. Communication Interfaces

- SPI: STM32 $\leftrightarrow$ LoRa SX1278 (PA4/PA5/PA6/PA7, PB0).
- I2C: STM32 $\leftrightarrow$ SSD1306 OLED display (PB6/PB7).
- UART2: STM32 $\leftrightarrow$ GPS NEO-6M (PA3 = RX).
- GPIO: Push buttons (input) and LED indicators (output).

V. HARDWARE SPECIFICATIONS

A. Component Overview

No.	Component	Specification	Role
1	STM32 Blue Pill	STM32F103C8T6, 72 MHz, 3.3 V	Main controller
2	LoRa SX1278	433 MHz, SPI, CSS modulation	Long-range TX/RX
3	OLED SSD1306	128×64 px, I2C interface	Display messages
4	GPS NEO-6M	UART, 9600 baud	Location tracking
5	Push Buttons	Tactile switch, GPIO	Message selection
6	LEDs + 220 Ω	5 mm LED, resistor	Status indication
7	Li-ion Battery	3.7 V rechargeable	Portable power
8	AMS1117-3.3 V	Voltage regulator	Stable 3.3 V supply

Table I: Hardware Components and Specifications

B. Key LoRa Parameters

The LoRa SX1278 is governed by four configurable parameters — Transmission Power (2–20 dBm), Spreading Factor (SF6–SF12), Bandwidth (125/250/500 kHz), and Coding Rate — enabling a data rate range of 0.3–50 kbps and a communication range of 5–10 km depending on terrain. Operating at 433 MHz ISM band provides superior penetration through dense environments compared to 868/915 MHz variants.

C. Electrical Specifications

- Microcontroller: STM32F103C8T6, Operating Voltage: 3.3 V.
- Modulation: Chirp Spread Spectrum (CSS).
- Display: 128×64 OLED, I2C interface.
- Power Supply: 3.7 V Li-ion battery with AMS1117-3.3 V regulation.
- Communication Range: Up to 5–10 km (terrain dependent).
- Transmission Mode: Half-duplex.

D. Mechanical Specifications

- Enclosure: Compact handheld acrylic / ABS plastic casing.
- Dimensions: Approx. 10 cm × 8 cm × 4 cm (customizable).
- Weight: Less than 500 grams.
- Portability: Designed for handheld field deployment.

VI. SOFTWARE SYSTEM DESIGN

A. Development Environment

- IDE: STM32CubeIDE / Arduino IDE.
- Language: Embedded C.
- Debugger: ST-Link / USB-Serial adapter.
- OS: Windows 10 / 11.

B. Transmitter Algorithm

The transmitter node operates on a continuous polling loop: (1) Initialize GPIO, SPI, and UART2; (2) Configure LoRa SX1278 in transmit mode; (3) Poll button inputs (PA0 = HELP, PB1 = SAFE); (4) On button press — debounce, identify message type, optionally read GPS NMEA data; (5) Encode and transmit via LoRa; (6) Activate corresponding LED; (7) Return to step 3.

C. Receiver Algorithm

The receiver node operates in a continuous listen-decode-display cycle: (1) Initialize GPIO, SPI, and I2C; (2) Configure LoRa SX1278 in receive mode; (3) Poll DIO0 for incoming packet interrupt; (4) On packet — verify checksum, extract payload, determine sender ID; (5) Clear OLED, format and display message; (6) Return to step 3.

VII. RESULTS AND PERFORMANCE EVALUATION

A. System Testing

The developed prototype was successfully tested for long-range wireless communication. Both nodes established reliable communication without cellular networks or internet connectivity. Emergency messages 'HELP' and 'SAFE' were transmitted from Node 1 and displayed accurately on the SSD1306 OLED at Node 2. The system demonstrated stable communication over considerable distances under NLOS conditions, confirming the effectiveness of CSS modulation in penetrating physical obstacles.

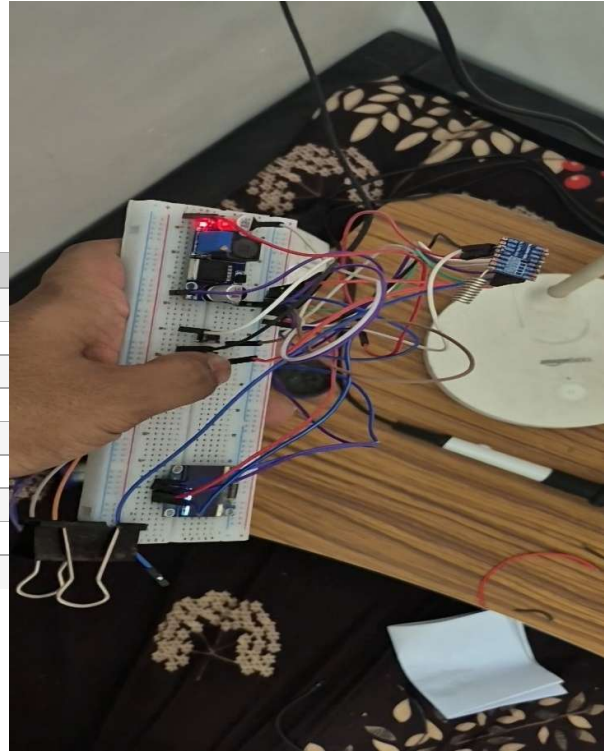
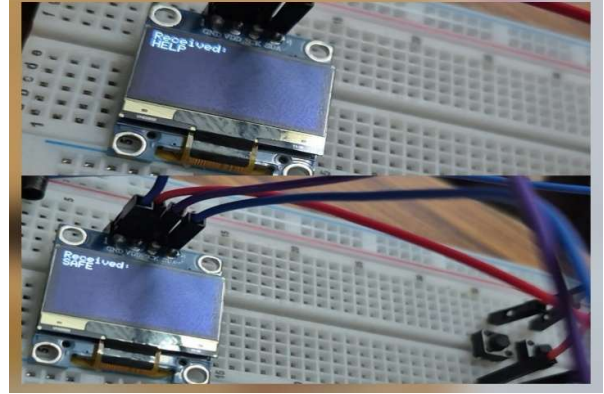
The GPS NEO-6M module successfully provided NMEA location data appended to transmitted messages. Power consumption was observed to be low, confirming suitability for battery-operated field applications.

B. Performance Summary

Parameter	Result
Operating Frequency	433 MHz (ISM Band)
Modulation	Chirp Spread Spectrum (CSS)
Communication Range	Up to 5–10 km
Data Rate	0.3–50 kbps (configurable)
NLOS Reliability	Stable — confirmed
Power Supply	3.7 V Li-ion battery
Operating Voltage	3.3 V (regulated)
GPS Integration	NEO-6M NMEA data
Total Project Cost	Approx. ₹2,558

Table II: System Performance Summary

C. Project Enclosure and Working



D. Bill of Materials

No.	Component	Spec	Qty	Rate (₹)	Total (₹)
1	STM32 Blue Pill	STM32F103C8T6	2	250	500
2	LoRa SX1278	433 MHz, SPI	2	450	900
3	OLED SSD1306	128×64, I2C	1	200	200
4	GPS NEO-6M	UART	1	300	300
5	Push Buttons	Tactile Switch	4	10	40
6	LEDs (5 mm)	Standard LED	4	5	20
7	Resistors 220 Ω	1/4 W	4	2	8
8	Li-ion Battery	3.7 V	2	150	300
9	Voltage Reg.	AMS1117-3.3 V	2	20	40
10	Jumper Wires	Set	1	100	100
11	PCB/Breadboard	General Purpose	1	150	150

Table III: Bill of Materials — Total ₹2,558

VIII. APPLICATIONS

- Disaster Management and Rescue Operations: Rapid deployment in earthquake, flood, or cyclone-affected zones.
- Remote and Rural Communication: Emergency communication in villages, forests, and mountains with no cellular coverage.
- Trekking and Expedition Safety: Portable distress-signaling device for hikers and mountaineers.
- Military and Defence: Covert, infrastructure-independent field communication.
- Forest and Wildlife Monitoring: Emergency alerts at ranger stations and monitoring outposts.
- Search and Rescue Coordination: Maintaining communication across large search areas.

IX. CONCLUSION AND FUTURE SCOPE

A. Conclusion

The proposed long-range, off-grid emergency alert messaging system successfully demonstrates reliable wireless communication using LoRa technology without requiring internet access or cellular networks. The integration of STM32F103C8T6 microcontrollers, LoRa SX1278 modules, an SSD1306 OLED display, GPS NEO-6M, and LED indicators provides a portable, cost-effective, and user-friendly solution for emergency remote communication. At a total hardware cost of approximately ₹2,558, the prototype is economically viable for mass production and large-scale deployment. The system is highly suitable for disaster management, trekking, rescue operations, and rural or forest environments.

B. Future Scope

- Mesh Network: Expanding to multi-node mesh networks for broader area coverage.
- Mobile App Integration: Android/iOS companion app via Bluetooth or Wi-Fi bridge.
- Real-Time GPS Mapping: Live location tracking with map visualization at the control centre.
- AES Encryption: Data encryption for security in sensitive deployments.
- Solar Charging: Integrating solar panels for sustainable self-powered field operation.
- LoRa WAN Gateway: Connecting to a LoRa WAN server for city-wide emergency infrastructure.

REFERENCES

- E. Senthilvel Murugan, M. Shunmuga Lakshmi and A. Ramya Lakshmi, 'Effective Communication System for Disaster Management Using LoRa,' [1] IJAEM, vol. 7, issue 05, pp. 208–214, May 2025.
- Q. Zhou, K. Zheng, L. Hou, J. Xing and R. Xu, 'Design and Implementation of Open LoRa for IoT,' [2] IEEE Access, vol. 7, pp. 100649–100657, 2019.
- S. Blesson, 'LoRa Based Emergency Communication System,' [3] ISJEM, vol. 03, pp. 1–9, 2024. DOI: 10.55041/ISJEM01628.
- D. Mahakal et al., 'Cordless Communication System using LoRa Technology,' [4] IJERT, vol. 14, issue 11, Nov. 2025. DOI: 10.17577/IJERTV14IS110335.
- G. Pasolini, C. Buratti et al., 'Smart City Pilot Projects Using LoRa and IEEE 802.15.4 Technologies,' [5] Sensors, vol. 18, p. 1118, 2018.
- F. Aoudia, M. Gautier et al., 'Long-short Range Communication Network Leveraging LoRa and Wake-up Receiver,' [6] Microprocessors and Microsystems, vol. 56, pp. 184–192, 2018.
- J. Petäjäjärvi et al., 'Performance of a Low-Power Wide-Area Network Based on LoRa Technology,' [7] Int J Distrib Sens N, vol. 13, 2017.
- Abhishek IJ, 'LoRa-Based Emergency Communication Device Without Network or External Power,' [8] ISE, Oxford College of Engineering, Bengaluru, India.